License Management User Guide

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6 Managing Licenses

Introduction

The Ixia Registration Utility (IRU) provides a means for monitoring and managing the various installed licenses on a per chassis or workstation basis. Before starting the license management process, it is helpful to have the following information available:

1. Registration Number and password for the license – this comes as an e-mail from Ixia to the Lab Manager and should have been forwarded to the person in charge of the installation. The password is necessary for operations performed on licenses from the License Management dialog.
2. Workstation or Ixia Chassis – this is the name or IP address of the computer or chassis that requires the license. This should also be provided by the Lab Manager.
3. License Server – **if** the lab will use a central License Server, the Lab manager should have provided its name or IP address.

License configuration options are also supported via a text file.

The following topics are covered in this chapter:

- *License Management* on page 6-2
- *Floating License Configuration Options* on page 6-17

License Management

License management can be done after the IRU is installed and running, and is able reach the various computers, chassis, and/or License Servers in the test environment. Make sure the network connectivity requirements discussed in *Installation Scenarios* are understood, and that the necessary information discussed above in *Introduction* is available.

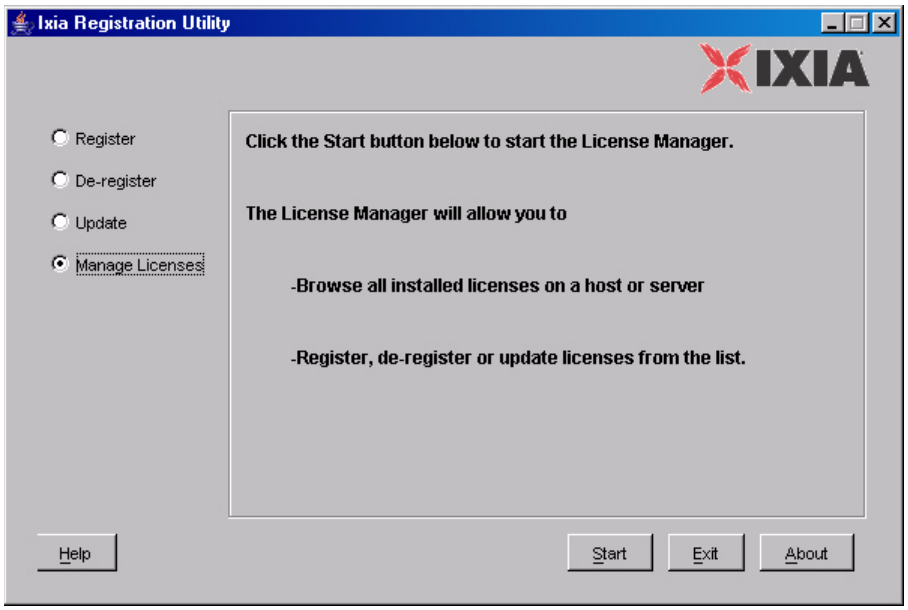
The following sections explain how to start and operate license management:

- *Opening the Ixia Registration Utility*
- *License Management Dialog*
- *License Menu Options*
- *Adding Licenses*
- *Moving Licenses*
- *Tools Menu Options*

Opening the Ixia Registration Utility

License management is started by opening the IRU. When the IRU is opened, the screen shown in Figure 6-1 is displayed. This is the main IRU screen that allows the user to select various registration operations. To open the IRU, select *Start* > *Programs* > *Ixia* > *Licensing* > *IRU* from the Start menu.

Figure 6-1. Ixia Registration Utility Main Dialog



- Select the *Manage Licenses* radio button.
- Select the *Start* button.

The other radio button operations are described in the following chapters:

- Registration – Chapter 2, Registering Licenses
- De-registration – Chapter 3, De-registering Licenses.
- Update – Chapter 4, Updating Licenses.

License Management Dialog

The main tool for managing local and remote licenses (such as those on a License Server) is the License Management dialog, shown in Figure 6-2.

Figure 6-2. Ixia Licence Management Utility Dialog

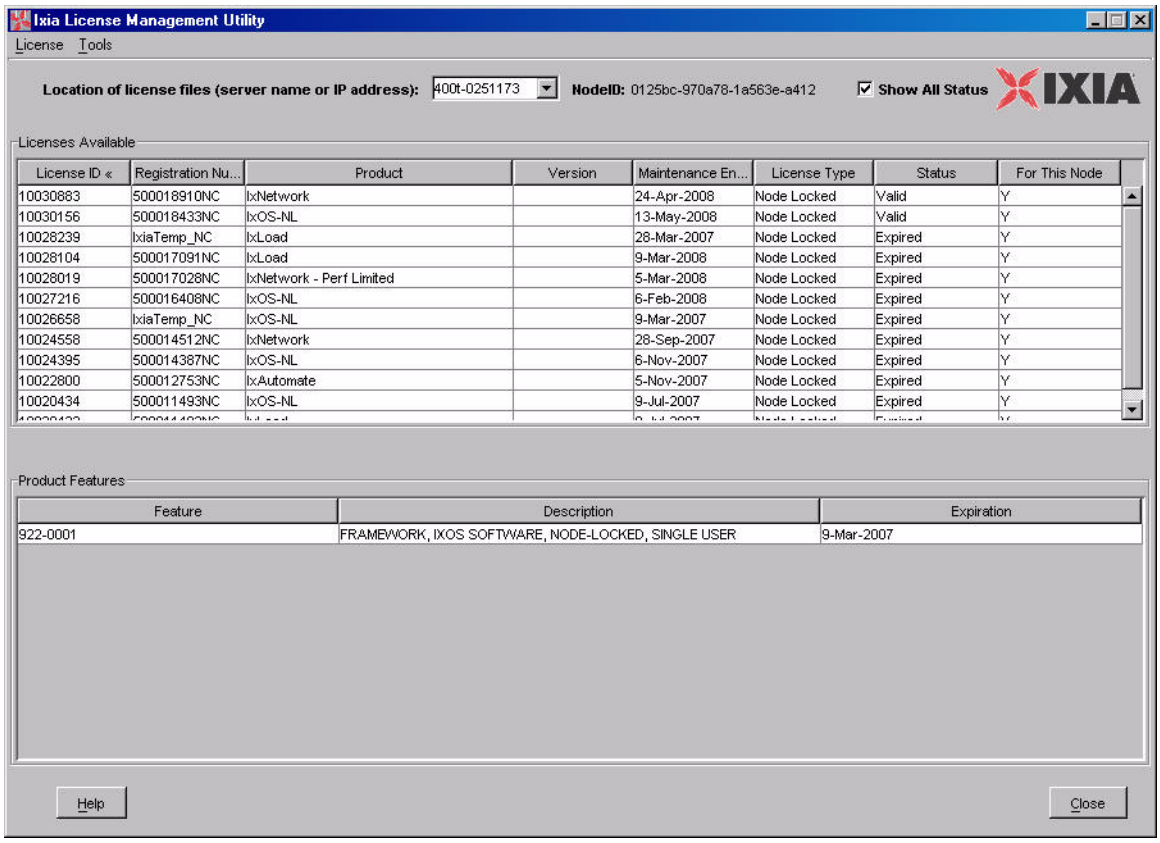


Table 6-1 describes the fields in the *Ixia License Management* dialog.

Table 6-1. Field Explanations

Section	Field	Description
Top of Dialog		
	License Menu Option	A menu list of registrations options for a selected license. These options are grayed out until a license is selected. The options in this menu are described in *License Menu Options* on page 6-6.

Table 6-1. Field Explanations

Section	Field	Description
	Location of license files	A pull down menu of the chassis or workstations that have been accessed by this particular IRU. Selecting a chassis or workstation from the list displays installed and available licenses. The chassis or workstation selected must be accessible from the local machine. The local machine is referred to in this list as “localhost”.
	Node ID	The assigned Node ID for the selected chassis or workstation. This number is assigned to the chassis or workstation by the IRU during the registration process to differentiate it from other nodes (chassis or workstations).
	Show All Status	Selecting this checkbox shows all the licenses on the specified Node, and their current status (either valid or expired).
Licenses Available		Displays the installed licenses for the selected chassis or workstation.
	License ID	The license filename (without the extension) of the selected license file.
	Registration Number	The registration number for the selected license. This number is provided by Ixia, along with its accompanying password, as described in What is License Management?
	Product	The product for which the license allows use.
	Version	The version of the licensed product. This field is valid mostly for licenses issued before Licensing 2.40. Either this field or the *Maintenance End Date* field has information for the current license.
	Maintenance End Date	The expiration date of the of the product maintenance. Once this date is passed, a new license is required. The end date is the same as the last day of the product’s service contract. This field is valid mostly for licenses issued after Licensing 2.40. Either this field or the *Version* field has information for the current license. Note that an existing date-based license that has expired can sometimes interfere with a valid evaluation license.

Table 6-1. Field Explanations

Section	Field	Description
	License Type	Whether the license is *node-locked* or *Floating*. Node-locked licenses only work for a single chassis or workstation (known as a “node”). Floating licenses can be used by any chassis or workstation in a test environment, up to a certain number.
	Status	The status of the license. Valid or Expired.
	For This Node	Indicates if the license is for a product that is on the node being viewed.
Product Features		Displays what features are available with the selected license.
	Feature	The name of the Ixia product that is licensed for use on the selected chassis or workstation.
	Description	A brief description of the product features.
	Expiration	When, if at all, the license for this product expires. If there is no expiration, this field reads *Permanent*.

License Menu Options

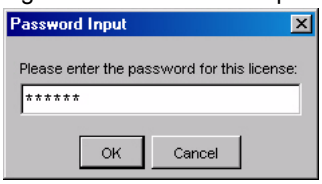
The *License* menu in the *Ixia License Management* dialog allows for more streamlined licensing operations. Using the options in this menu allow the user to skip the step of entering the Registration Number and Password.

The options in the *License* menu are:

- De-register – Allows for de-registering the license. Selecting this option starts the de-registration process which is described in Chapter 3, De-registering Licenses.
- Update – Allows for updating a selected license. Selecting this option starts the update process, which is described in Chapter 4, Updating Licenses.
- Add License – Allows for adding a previously registered license to its workstation, chassis, or server. Selecting this option starts the add license process, which is described in *Adding Licenses* on page 6-7.

The de-register and update operations require the password provided with the registration number, as described in *Introduction* above. Once one of these operations is selected, the *Password Input* dialog is displayed, as shown in Figure 6-3.

Figure 6-3. Password Input Dialog



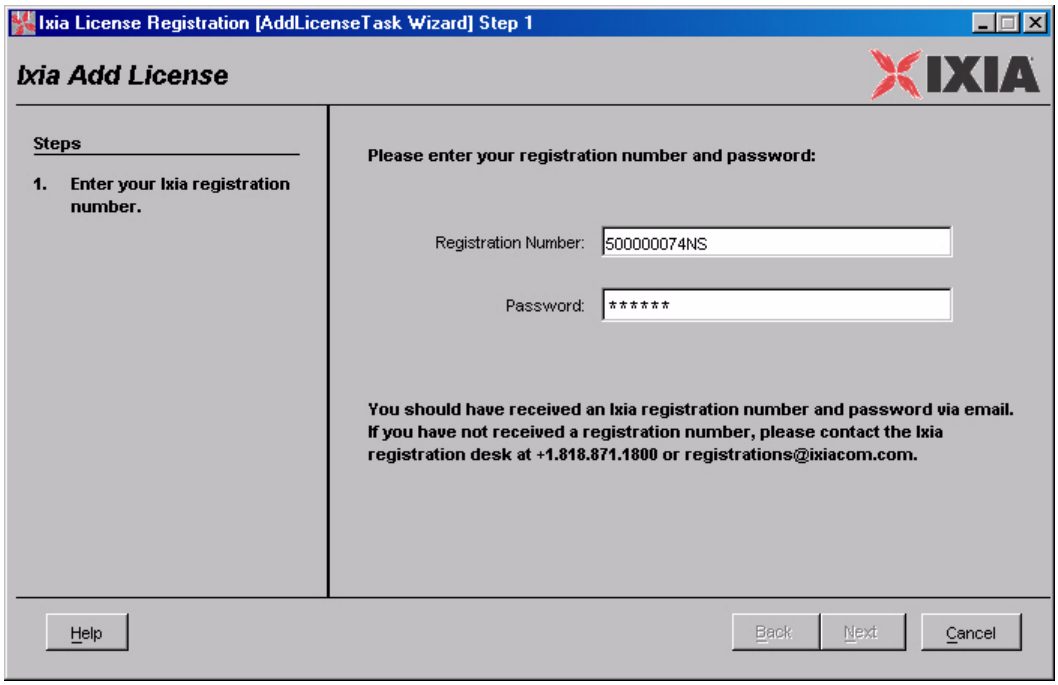
The process for adding a license is described in *Adding Licenses* on page 6-7 below.

Adding Licenses

One of the features in the LMU *License* menu is *Add License*. A license can be added to a workstation or chassis after it has been downloaded from the Ixia's Registration Server, but for some reason was not installed. Most times this occurs when during offline registration, the user closes the Ixia Registration Utility (IRU) before completing *Step 8: Online – Save the License File*.

1. From the LMU *License* menu, select *Add License*, or press the *Ctrl + A* keys. The dialog shown in Figure 6-4 is displayed.

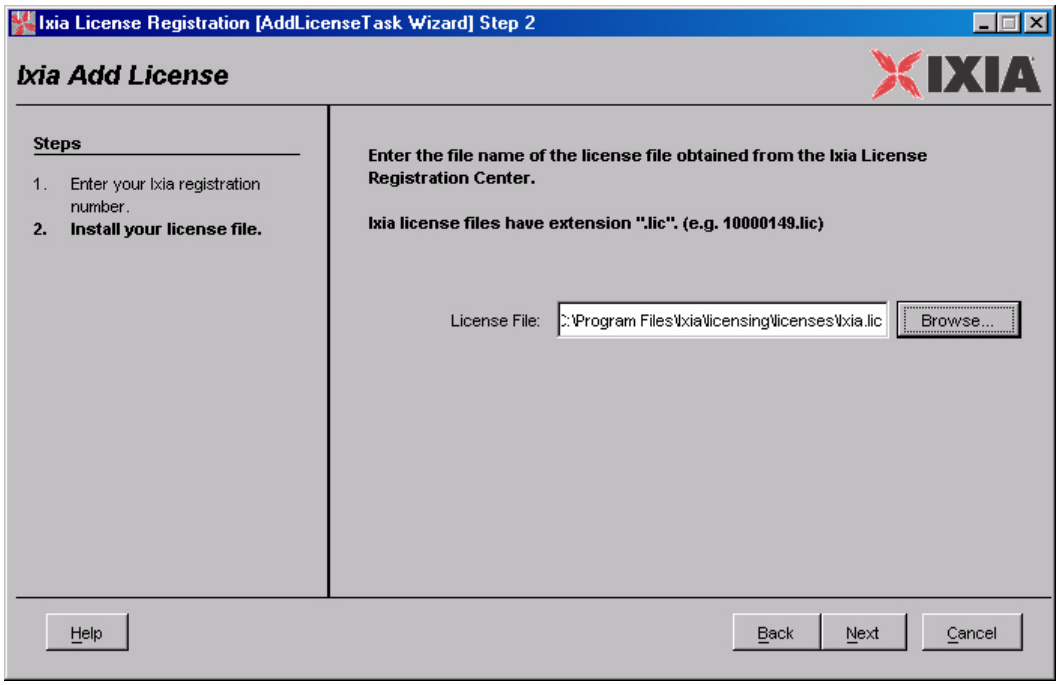
Figure 6-4. Add License – Step 1



Enter the Registration Number and Password for the license being added, then select the *Next* button.

2. The next dialog displayed is shown in Figure 6-5.

Figure 6-5. Add License – Step 2

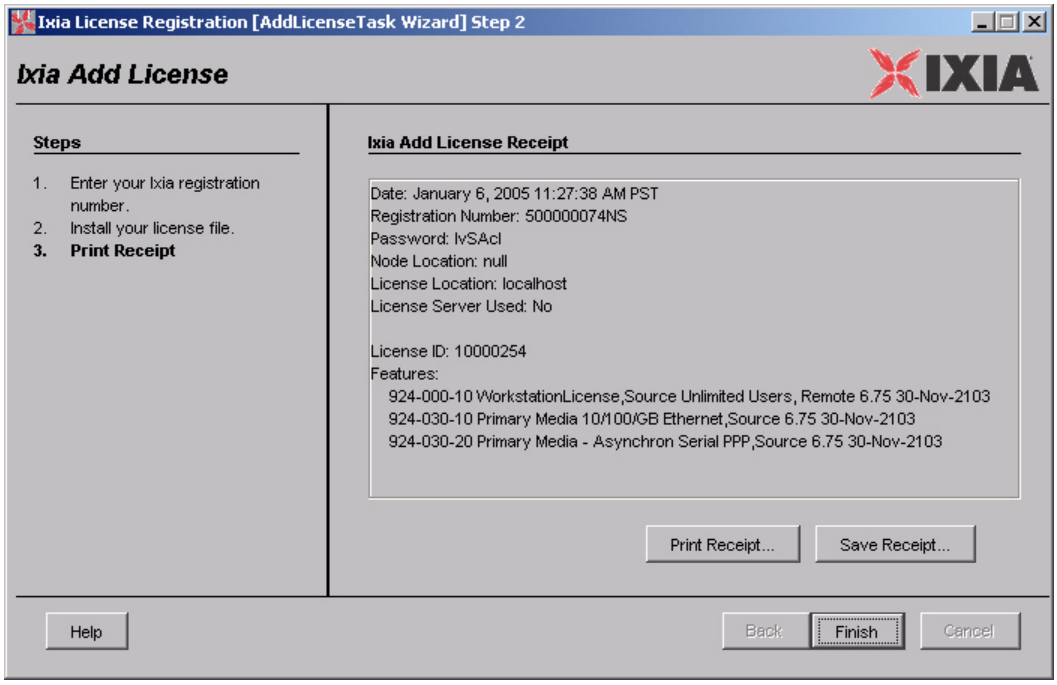


Enter the file license file name in the field provided, or use the Browse button to locate it. Once selected, the path to the license should display in the field. Press the *Next* button.

Note. The machine running the add license operation must be connected to the chassis or workstation to which the license has been assigned.

3. The process should install the license the license to the correct chassis or workstation. When it is finished the dialog shown in Figure 6-6 is displayed.

Figure 6-6. Add License – Step 3



To save the receipt to disk, press the *Save Receipt...* button and designate a file. To make a printout of the receipt, press the *Print Receipt...* button and select a printer.

Press *Finish* to exit the Add License process.

Moving Licenses

Licenses can be moved between chassis, workstations, and License Servers. To move a license from one machine to another:

1. De-register the license from the original chassis or workstation, as described in Chapter 3, De-registering Licenses.
2. Register the license to the new machine, as described in Chapter 2, Registering Licenses.

Tools Menu Options

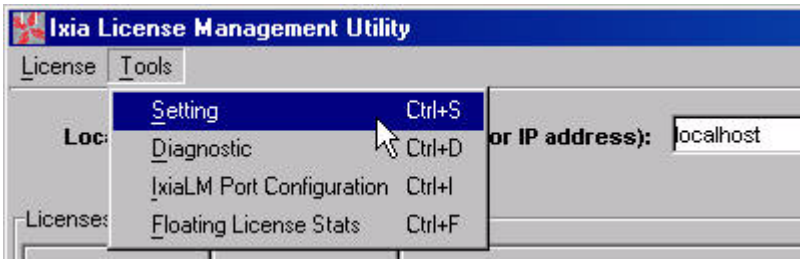
The *Tools* menu options are listed below:

- *Setting*
- *Using the Diagnostic Utility*
- *IxiaLM Port Configuration*
- *Floating License Traffic Statistics*

Setting

The *Setting* control allows the user to specify a proxy server registration and to force the IRU to use offline registration. To use the *Setting* control, select *Tools*> *Setting* in the menu bar, as shown in Figure 6-7 on page 6-10.

Figure 6-7. Open the Setting Dialog



There are two tab options in the *Setting* dialog:

- *Proxy* on page 6-11
- *Force Off-Line* on page 6-12

Proxy

The *Proxy* tab page controls the use of proxy server in license registration. If a proxy server is used in the network from which the registration is being performed, it is necessary to set the name and port of the proxy server.

Note. Normal Internet Explorer proxy settings are recognized by Licensing, without setting the proxy options.

If there is a problem with Licensing due to proxy settings that cannot be remedied using these controls, it is best to use offline registration.

The *Proxy* tab page is show in Figure 6-8 on page 6-11.

Figure 6-8. Proxy Tab

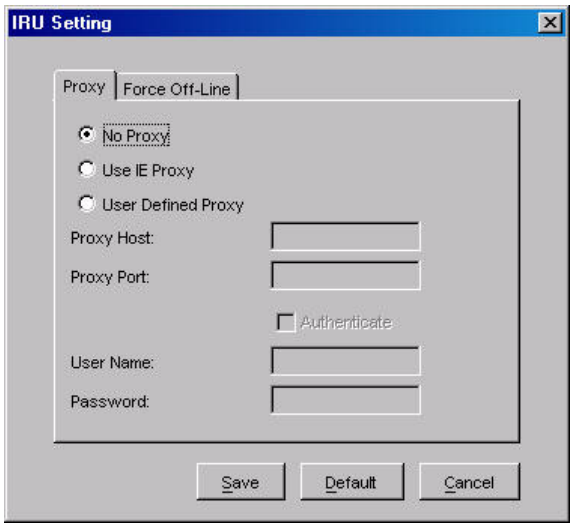


Table 6-2 on page 6-11 describes the functionality of the *Proxy* tab page.

Table 6-2. Proxy Tab Page Configuration

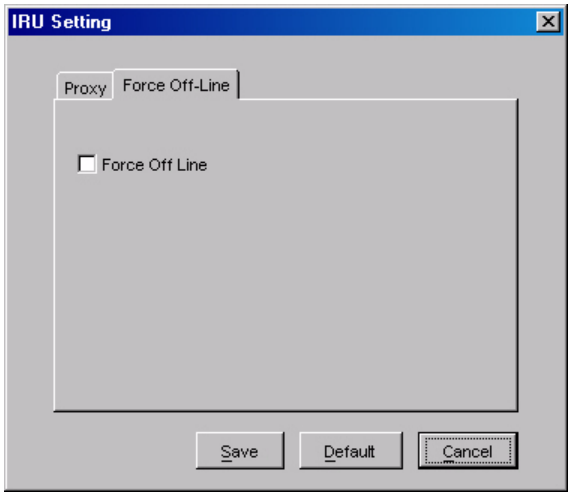
Control	Description
No Proxy	No proxy server is used to connect to the Internet. This is the default setting.
Use IE Proxy	Selecting this checkbox enables use of the Internet Explorer proxy settings.
Use Proxy	Selecting this checkbox enables the proxy server settings. The IRU will look for the specified proxy machine on the specified port.
Proxy Host	Enter the Host Name of the proxy server.
Proxy Port	Enter the UDP port number of the proxy server.
Authenticate	Selecting this checkbox enables proxy server user authentication settings.
User Name	The authorized user for the proxy server.
Password	The authorized user's proxy server password.

Force Off-Line

The *Force Off-Line* tab page controls whether the IRU forces an off-line registration or not.

Figure 6-9 on page 6-12 shows the *Force Off-Line* tab page.

Figure 6-9. Forced Off-Line Tab

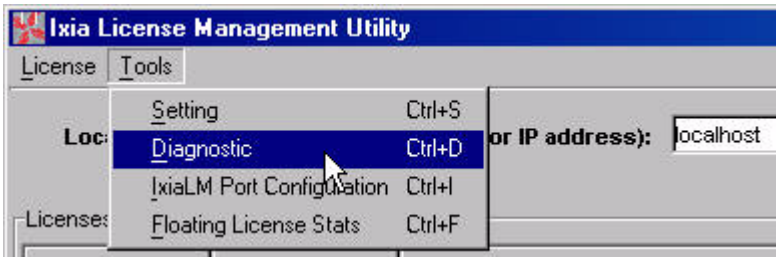


Select the *Force Off Line* checkbox to force the IRU to register licenses offline.

Using the Diagnostic Utility

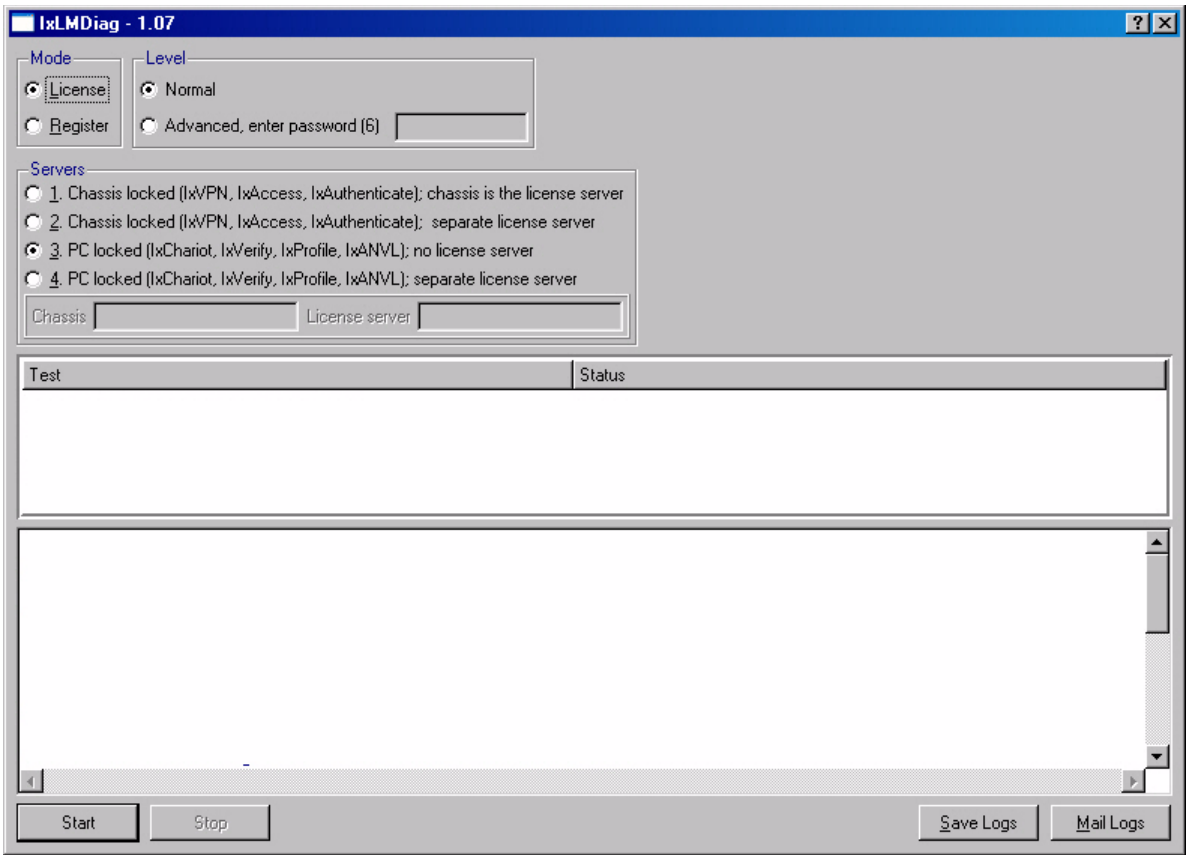
The diagnostic utility helps the user to troubleshoot and understand licensing installation and registration problems. To open this utility, select *Tools* > *Diagnostic* in the menu bar, as shown in Figure 6-10 on page 6-13:

Figure 6-10. Open the Diagnostic Utility



The diagnostic utility main window appears as in Figure 6-11 on page 6-13:

Figure 6-11. Diagnostic Utility



For specific information on using this utility, see Appendix C, Licensing Diagnostic Utility.

IxiaLM Port Configuration

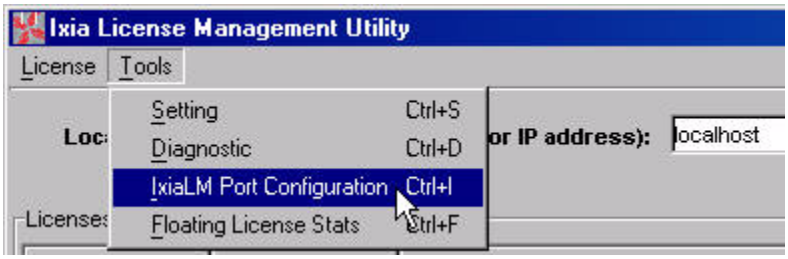
If the license server and the licensed node are separated by a firewall, the license server may have problems validating the node for licensed use. In this situation it may be necessary to specify a TCP port that works with the firewall setup on the localhost for the license

.

Note. This feature only affects the licenses on the localhost.

To open this utility, select *Tools* > *IxiaLM Port Configuration* in the menu bar, as shown in Figure 6-12 on page 6-14:

Figure 6-12. Open IxiaLM Port Configuration



The IxiaLM Port Configuration dialog appears as in Figure 6-13 on page 6-14:

Figure 6-13. IxiaLM Port Configuration Dialog

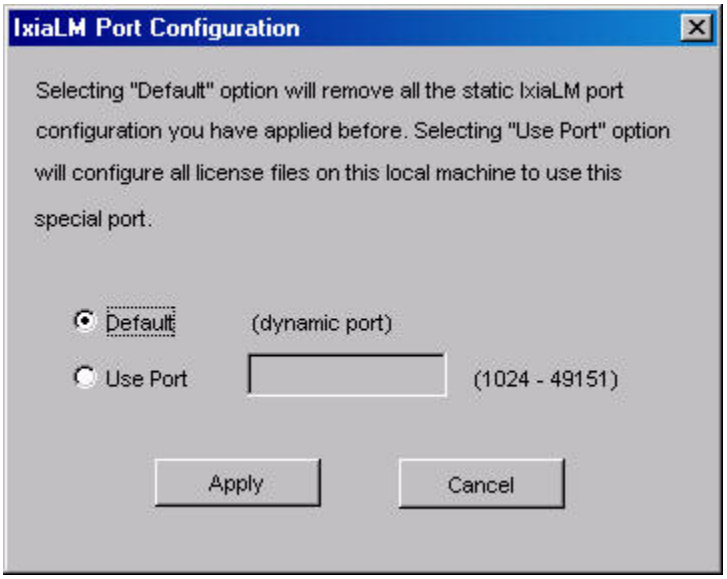


Table 6-3 explains the options in the IxiaLM Port Configuration dialog.

Table 6-3. Port Configuration Controls

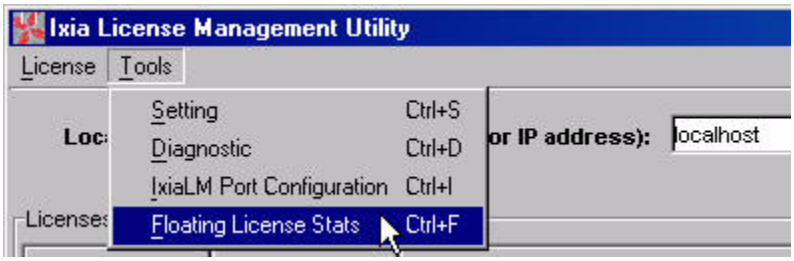
Control	Description
Default	(default) Selecting this radio button causes the licensing service to use the first available TCP port for node validation. This option overrides any previously configured port setting.
Use Port	Selecting this radio button cause the licensing service to use the specified TCP port for node validation. Valid TCP ports are 1024 through 49151.

Floating License Traffic Statistics

The Floating License Traffic Statistics feature provides a way for users to examine floating license statistics such as the total license token count, available token count, users that have checked out license tokens, and the number of license tokens each user has checked out.

To open this utility, select *Tools* > *Floating License Stats* in the menu bar, as shown in Figure 6-14 on page 6-15:

Figure 6-14. Open Floating License Stats



The Floating License Statistics dialog appears as in Figure 6-15 on page 6-16:

Figure 6-15. Floating License Statistics Dialog

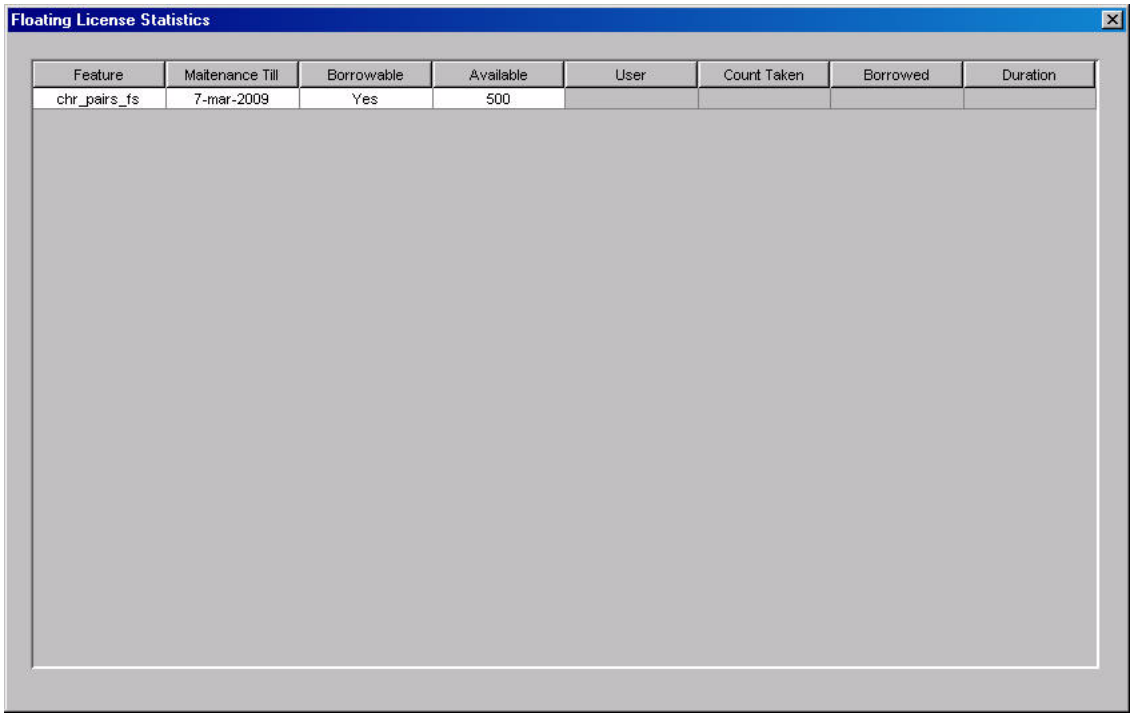


Table 6-4 explains the options in the Floating License Statistics dialog.

Table 6-4. Floating License Statistics Fields

Field	Description
Feature	Describes the license feature.
Maintenance Until	Shows the end maintenance date of the license.
Borrowable	Shows if the license can be borrowed.
Available	Shows the number of users that can access the license at the same time.
User	The user that has the license checked out.
Count Taken	Shows the number of ???
Borrowed	Shows whether the license is borrowed or not. Borrowed licenses are checked out for a specific amount of time.
Duration	If the license is borrowed, the length of time the license will be checked out.

Floating License Configuration Options

Ixia's licensing system supports configurable options for floating licenses by installing an option file, called `ixialm.opt`, with the software.

A floating license server administrator can control license usage based on user, host, domain, or other criteria, introducing a option file on the floating license server with some predefined option syntax. The administrator can modify the sample options syntax to achieve better administrative control on the floating license server.

The following options are supported:.

- Allow the use of features
- Deny the use of features
- Reserve licenses
- Restrict the number of licenses available
- Control the amount of information logged about license usage
- Enable a report log file

Initially, all options are disabled.

Syntax

The following sections detail the file options syntax and how they are used. The syntax is used in the ixialm.opt file, and is modified with any text editor. In the following command syntax:

- Text in capital letters is required, exactly as shown
- Text surrounded by open and closed carrot symbols (< and >) is optional
- Italicized text denotes a variable and must be replaced with some meaningful value.

BORROW_LOWWATER

```
BORROW_LOWWATER ixia_feature integer
```

Sets the number of BORROW licenses that cannot be borrowed. This number sets the minimum number of free licenses. For example, if this variable is set to 10, and a server has twenty licenses, only 10 can be borrowed at any one time.

Example:

```
BORROW_LOWWATER IXNETWORK_5.30 10
```

DEBUGLOG

```
DEBUGLOG <+> filename
```

Writes debug log information for this vendor daemon to the specified file. The addition of a plus sign (+) before the logfile name appends the log. If no plus sign is used, restarting the license server service overwrites the previous log.

Example:

```
DEBUGLOG + myixlmlog.log
DEBUGLOG newixlmlog.log
```

EXCLUDE

```
EXCLUDE ixia_feature exclude_level value
```

Deny a user or users access to a feature, based on specified parameters. The exclusions parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
EXCLUDE IXNETWORK_5.30 USER michael
EXCLUDE IXNETWORK_5.30 HOST MICHAEL-PC1
EXCLUDE IXNETWORK_5.30 DISPLAY MICHAEL-PC1
EXCLUDE IXNETWORK_5.30 INTERNET 10.205.20.67
EXCLUDE IXNETWORK_5.30 INTERNET 10.205.20.*
EXCLUDE IXNETWORK_5.30 GROUP WRITERS
EXCLUDE IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

EXCLUDE_BORROW

```
EXCLUDE_BORROW ixia_feature exclude_level value
```

Deny a user or users the ability to borrow BORROW licenses, based on specified parameters. The exclusions parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
EXCLUDE_BORROW IXNETWORK_5.30 USER michael
EXCLUDE_BORROW IXNETWORK_5.30 HOST MICHAEL-PC1
EXCLUDE_BORROW IXNETWORK_5.30 DISPLAY MICHAEL-PC1
EXCLUDE_BORROW IXNETWORK_5.30 INTERNET 10.205.20.67
EXCLUDE_BORROW IXNETWORK_5.30 INTERNET 10.205.20.*
EXCLUDE_BORROW IXNETWORK_5.30 GROUP WRITERS
EXCLUDE_BORROW IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

EXCLUDEALL

```
EXCLUDEALL ixia_feature exclude_level value
```

Deny a user or users to all features, based on specified parameters. The exclusions parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in

the case of a terminal server environment, the terminal server client name. Display names are case sensitive.

- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
EXCLUDEALL IXNETWORK_5.30 USER michael
EXCLUDEALL IXNETWORK_5.30 HOST MICHAEL-PC1
EXCLUDEALL IXNETWORK_5.30 DISPLAY MICHAEL-PC1
EXCLUDEALL IXNETWORK_5.30 INTERNET 10.205.20.67
EXCLUDEALL IXNETWORK_5.30 INTERNET 10.205.20.*
EXCLUDEALL IXNETWORK_5.30 GROUP WRITERS
EXCLUDEALL IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

GROUP

```
GROUP MYGRP user1 <user2>...<userN>
```

Defines a group of users for use with the options statements.

Example:

```
GROUP MYGRP john michael doug
```

GROUPCASEINSENSITIVE

```
GROUPCASEINSENSITIVE ON
```

Turns off case sensitivity for user and host lists specified in GROUP and HOST_GROUP keywords. By default, the license server is case sensitive.

Example:

```
GROUPCASEINSENSITIVE ON
```

HOST_GROUP

```
HOST_GROUP MYHGRP host1 <host2>...<hostN>
```

Defines a group of hosts for use with the options statements.

Example:

```
HOST_GROUP MYHGRP MICHAEL-LAP2 MICHAEL-PC2 JOHN-WORK-PC
```

INCLUDE

```
INCLUDE ixia_feature include_level value
```

Allow a user or users access to a feature, based on specified parameters. The inclusion parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
INCLUDE IXNETWORK_5.30 USER michael
INCLUDE IXNETWORK_5.30 HOST MICHAEL-PC1
INCLUDE IXNETWORK_5.30 DISPLAY MICHAEL-PC1
INCLUDE IXNETWORK_5.30 INTERNET 10.205.20.67
INCLUDE IXNETWORK_5.30 INTERNET 10.205.20.*
INCLUDE IXNETWORK_5.30 GROUP WRITERS
INCLUDE IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

INCLUDE_BORROW

```
INCLUDE_BORROW ixia_feature include_level value
```

Allow a user or users access to borrow a feature, based on specified parameters. The inclusion parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
INCLUDE_BORROW IXNETWORK_5.30 USER michael
INCLUDE_BORROW IXNETWORK_5.30 HOST MICHAEL-PC1
INCLUDE_BORROW IXNETWORK_5.30 DISPLAY MICHAEL-PC1
INCLUDE_BORROW IXNETWORK_5.30 INTERNET 10.205.20.67
INCLUDE_BORROW IXNETWORK_5.30 INTERNET 10.205.20.*
INCLUDE_BORROW IXNETWORK_5.30 GROUP WRITERS
INCLUDE_BORROW IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

INCLUDEALL

```
INCLUDEALL ixia_feature include_level value
```

Allow a user or users access to all features, based on specified parameters. The inclusion parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in

the case of a terminal server environment, the terminal server client name. Display names are case sensitive.

- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
INCLUDEALL IXNETWORK_5.30 USER michael
INCLUDEALL IXNETWORK_5.30 HOST MICHAEL-PC1
INCLUDEALL IXNETWORK_5.30 DISPLAY MICHAEL-PC1
INCLUDEALL IXNETWORK_5.30 INTERNET 10.205.20.67
INCLUDEALL IXNETWORK_5.30 INTERNET 10.205.20.*
INCLUDEALL IXNETWORK_5.30 GROUP WRITERS
INCLUDEALL IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

MAX

```
MAX max_users ixia_feature include_level value
```

Limit usage for a particular feature/group-prioritizes usage among users. The inclusion parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
MAX 5 IXIA_FEATURE1 USER michael
MAX 5 IXIA_FEATURE1 HOST MICHAEL-PC1
MAX 5 IXIA_FEATURE1 DISPLAY MICHAEL-PC1
MAX 5 IXIA_FEATURE1 INTERNET 10.205.20.67
MAX 5 IXIA_FEATURE1 INTERNET 10.205.20.*
MAX 5 IXIA_FEATURE1 GROUP MYGRP
MAX 5 IXIA_FEATURE1 HOST_GROUP MYHGRP
```

MAX_BORROW_HOURS

```
MAX_BORROW_HOURS ixia_feature value
```

Changes the maximum borrow period for the specified feature.

Example:

```
MAX_BORROW_HOURS IXNETWORK_5.30 36
```

NOLOG

```
NOLOG value
```

Turns off logging of certain items in the debug log file. The options are:

- IN – log in
- OUT – log out
- DENIED – denied log in

Example:

```
NOLOG IN
NOLOG OUT
NOLOG DENIED
```

REPORTLOG

Specify that a report log file suitable for use by the FLEXnet Manager license usage reporting tool be written.

RESERVE

```
RESERVE ixia_feature include_level value
```

Reserves licenses for a user or group of users/hosts. The reserve parameter is one of the following:

- USER – reserves for a specific user. User names are case sensitive.
- HOST – reserves for a specific host machine. Host names are case sensitive.
- DISPLAY – includes the display machine. On UNIX, DISPLAY is /dev/ttyxx (which is always /dev/tty when an application is run in theback-ground) or the X-Display name. On Windows, it is the system name or, in the case of a terminal server environment, the terminal server client name. Display names are case sensitive.
- INTERNET – reserves for a specific IP address or IP subnet. Wildcard characters can be used in the IP address.
- GROUP – reserves for a pre-defined group of users (see *GROUP* on page 6-20 for more information on defining a user group). Group names are case sensitive.
- HOST_GROUP – reserves for a pre-defined group of hosts (see *HOST_GROUP* on page 6-21 for more information on defining a host group). Host group names are case sensitive.

Examples:

```
RESERVE IXNETWORK_5.30 USER michael
RESERVE IXNETWORK_5.30 HOST MICHAEL-PC1
RESERVE IXNETWORK_5.30 DISPLAY MICHAEL-PC1
RESERVE IXNETWORK_5.30 INTERNET 10.205.20.67
RESERVE IXNETWORK_5.30 INTERNET 10.205.20.*
RESERVE IXNETWORK_5.30 GROUP WRITERS
RESERVE IXNETWORK_5.30 HOST_GROUP WRITERS-PCS
```

TIMEOUT

```
TIMEOUT ixia_feature value
```

Specifies the idle timeout for a feature. After the set amount of time passes without activity on the license, it is returned to the free pool for use by another user.

Example:

```
TIMEOUT IXIA_FEATURE1 3600
```

TIMEOUTALL

```
TIMEOUTALL value
```

Set timeout on all features. If this is set below 900, the timeout is 900 seconds.

Example:

```
TIMEOUTALL 3600
```

B Ixia Registration Utility (IRU) Errors and Troubleshooting

This appendix provides basic licensing troubleshooting information, and lists various errors that can occur during the registration, de-registration, or update process.

The following sections cover the troubleshooting and error messages for licensing:

- *General Problems and Troubleshooting* on page B-2
- *Error Messages* on page B-8

General Problems and Troubleshooting

The following sections outlines some common issues users may encounter when dealing with the Ixia licensing software. This section is not meant to be all inclusive.

For more detailed instruction for coping with licensing issues, see the FAQ and troubleshooting guide on the licensing section of the Ixia website.

Proxy Servers and Firewalls

If licenses are installed on a network that uses either proxy servers or firewalls, it may be necessary to do an offline registration, even if the Installation Host can reach the Internet.

Follow the procedures for offline registration, de-registration, and update provided in the following sections:

- *Offline License Registration – Step by Step* on page 2-10
- *Offline License De-registration – Step by Step* on page 3-10
- *Offline License Update – Step by Step* on page 4-10

Note. If the system to which the license is being installed uses SOCKS version 5, it is not necessary to configure special proxy settings.

Linux Firewall

Some applications that run on the Linux operating system require that the Firewall be turned off for licensing to work correctly.

To disable the Linux Firewall, enter one of the following commands:

```
service iptables stop
```

or

```
/etc/init.d/iptables stop
```

Versions of Linux Supported

Ixia licensing currently supports Red Hat Linux version 9.0. Other versions of Linux are not fully supported.

Licensing Services

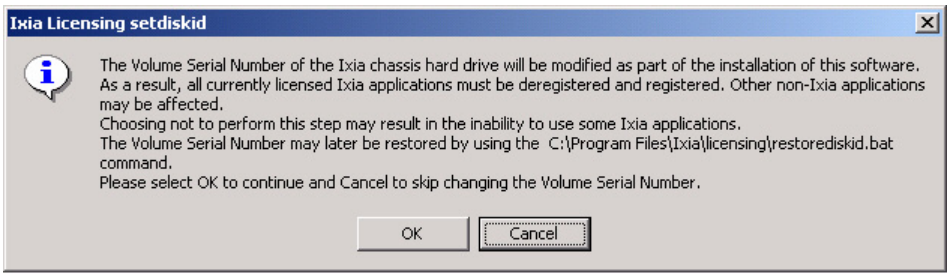
Once licensing is installed, the license server and Node ID generation run as Windows services. If there is a problem licensing or using a product, first check to see that the license services are running. These services are:

- IxiaLicenseServer
- noded

Disk ID Reset Utility

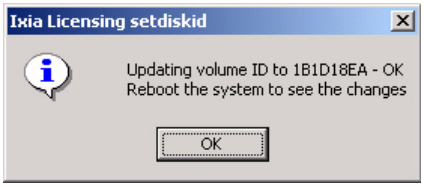
The utility is run at the time of the license installation. A dialog similar to the one shown in Figure B-1 is displayed:

Figure B-1. Set Disk ID Warning



Select *OK* to continue with the change in the disk ID number, select *Cancel* to retain the old disk ID. When finished, a dialog similar to figure is shown.

Figure B-2. Set Disk ID Complete



Select *OK* to complete the process.

The old disk ID number is stored in a file called `restorediskid.bat`. By running this batch file, the user can restore the old disk ID.

> **Note**. Changing the disk ID requires that you re-register all licensed applications on the chassis. To assist in this process, a script is included with Licensing that de-registers and re-registers all existing Ixia licenses automatically. The utility is named `UpdateUtil.exe` and is located at *C:\Program Files\Ixia\Licensing*.
>
> This utility is automatically run if the Disk ID is reset.

License Not Found Error

Most Ixia products require a license to function. When a valid license cannot be found, the Licensing software will generate an error message. I

First, check the IRU and run the diagnostics application, as described in Appendix C, *Licensing Diagnostic Utility*.

If the IRU hangs during the diagnostics application and does not complete:

1. Open a command window using the Start menu.
2. At the prompt enter the following command:

   `netstat -an.`

3. Check to see if ports 4501 and 27000 – 27009 are in a LISTENING. state. If not, investigate if these ports could be blocked on the chassis, PC, or in the network along the traffic path.
4. Open a web browser and enter the following URL:

   `http://localhost:4501`

   If the code text is returned, the port is open. If an error page is returned, the port is blocked.

5. Put the chassis IP in exceptions of proxy server.

These efforts should make the license server work. Make sure that IxExplorer‘s proxy settings are set to *Auto-detect Proxy* and *Bypass For Local Addresses*.

It is best to run IxExplorer first before other applications, to establish the connections to the licensing software.

Notes. Prior to IRU version 2.60, ports 4501 and 27000 - 2700 were required to be open. If these ports are blocked then you must open them on the chassis, client PC, or network so the client PC can communicate with the IRU software (even if the license resides in the IRU on the client).

With IRU version 2.60 and higher, the customer can manually select which ports are desired for the licensing traffic to communicate on. See *IxiaLM Port Configuration* on page 6-14 for additional details.

Other common causes can be:

- **Changing the chassis name (the default is the SN)**. In this case, change the name of the chassis back to the name that the license was registered under.
- **Manually moving the license from one node to another**. A forced deregistration (done by Ixia technical support) may be required if the license on the node doesn't match the one in the Ixia License Database.
- **Changing the system date**. Since many files will have a timestamp change by the system, changing the date back won’t resolve the problem. Often the solution requires all the changed files to be manually “touched” with the correct date. In the case where the number of files exceeds the allowed time, the only resolution is to reformat the hard-drive. For a chassis, this requires a “ghosting” procedure. For a PC, the customer must use another PC, or have the IT dept reformat the hard-drive.

IP Port Assignments on Ixia Chassis and Linux Port CPUs

The following table lists the services assigned to IP ports on Ixia chassis and port CPUs as of January 30, 2007. Services listed in the *Used on Chassis* column are accessed through the chassis management port (the NIC located in the rear of the chassis). Services listed in the *Used on Port CPU* column are accessed through the test ports

This information applies to all Ixia chassis and all load modules with port CPUs
.

Note. Do not expose any port on an Ixia chassis to an untrusted network.

Connections are initiated in the following directions:

Table B-1. Ixia Port Connections

Port	Direction
2809 – 2825	Bidirectional between client and chassis.
All others	From client to chassis.

Table B-2. List of Port Services

Port	Use	TCP	UDP	Used on Chassis	Used on Port CPU	IxOS 4.00	IxOS 5.10
9	Discard service	X	X		X	X	X
21	FTP daemon	X				X	X
22	SSH (IxLoad)	X		X			
23	Telnet daemon	X			X	X	X
58	Reserved	X			X	X	X
80	HTTP, License Management	X			X	X	X
103	PIM daemon (test port only)	X	X	X		X	X
111	Sun RPC portmap	X	X	X		X	X
123	NTP		X	X		X	X
125	Private NTP		X	X		X	X
135	Windows COM Services	X	X	X		X	X
179	BGP daemon	X			X	X	X
445	Windows COM services	X	X	X		X	X
797 – 800	NFS mounts		X		X	X	X

Table B-2. List of Port Services

Port	Use	TCP	UDP	Used on Chassis	Used on Port CPU	IxOS 4.00	IxOS 5.10
1024 – ?	Windows MStask.exe (Google for “mstask vulnerability” for details)	X	X	X		X	X
1080	SOCKS proxy	X		X		X	X
2048	Service management	X		X	X		
2049	NFS	X	X	X	X	X	X
2050 – 2111	Service management	X		X	X		
2600 – 2699	Ixia reserved	X			X	X	X
2705	IxVPN	X			X	X	X
2782	CliX	X		X		X	X
2809 – 2825	Aptixia CORBA	X		X		X	X
3600 – 3999	Ixia reserved	X			X	X	X
3705	IxVPN	X			X	X	X
4501	License Management	X		X		X	X
4555	IxTclServer	X		X		X	X
4900	VNC	X		X		X	X
5285	IxServer status connection to IxDodServer	X		X		X	X
5286	IxServer connection to Admin	X		X			X
5555	IxVPN	X			X	X	X
6001	Service manager	X		X	X	X	X
6002	Capture service	X			X	X	X
6003	Capture service relay	X		X		X	X
6004	Download on Demand broadcast data		X	X	X	X	X
6005	Download on Demand server	X		X		X	X
6101	IxNetwork	X		X	X		
6809 – 25	Aptixia/CORBA	X			X	X	X
6665	InterfaceManager (IxAuthenticate and IxAccess)	X			X	X	X

Table B-2. List of Port Services

Port	Use	TCP	UDP	Used on Chassis	Used on Port CPU	IxOS 4.00	IxOS 5.10
8003	IxAuthenticate	X			X	X	X
8008	IxNetwork	X			X	X	X
9101 – 2	Statistics engine (Statengine)	X		X	X	X	X
9613 – 9676	Service management	X		X	X	X	X
10115	IxChariot	X		X	X	X	X
17668	IxServer	X		X			X
17669 (4505h)	IxServer	X		X		X	X
17670 (4506h)	Statistics watch	X		X		X	X
17671 (4507h)	Protocol watch	X		X		X	X
17672 (4508h)		X		X			
17673	IxExplorer						
17777 (4571h)		X		X		X	X
26999 – 27009	License Management pre-2.40SP1 (if license server is running on chassis)	X		X		X	X
32769	mountd	X		X		X	X
32800 – 16	IxVPN CORBA	X			X	X	X
54321	IxVPN	X		X		X	X

Error Messages

The following section lists common error messages that the user may encounter while using the Ixia licensing system. There are two tables:

- *Registration Server Error Codes* on page B-8 – Lists problems or conditions that involve the Ixia Registration Server.
- *IRU and noded Error Codes* on page B-9 – Lists problems or conditions that involve the installation process on the customer site.

The tables also provide brief troubleshooting notes.

Table B-3. Registration Server Error Codes

Error Number	Description	Action
1001	Invalid Registration Number or Password	Please verify the registration number and password from the email sent from Ixia and try again. Either the registration number or password entered is incorrect.
1002	License Already in Use	A license has already been issued for this registration. Please de-register before trying to register again. There can be no more than one active license for each registration.
1003	Internal Server Error	A severe problem occurred with the Ixia Registration server. Please try again later.
1004	Invalid Deregistration Code	The de-registration code is incorrect. The user has made a mistake while copying the de-registration code from the LRCU to the website. Please provide the correct de-registration code. If this error has occurred for online registration then the keys are out of sync on the LRCU and the registration server.
1006	Registration is No Longer Valid	This registration number can no longer be used. The registration was invalidated manually buy an authorized Ixia personnel because of RMA or incorrect data entry. Please contact Ixia to get the new registration number.
1010	This registration is already in use with an older version of the software	This registration is already in use with an older version of the software. The user is has not de-registration from the Legacy Registration Server and is trying to register the new version. Please de-register the old version and try again.
1012	Invalid Data Format: Node ID should have the format xxxxxx-xxxxxx-xxxxxx-xxxx	The Node ID was incorrectly entered. Try cutting and pasting the Node ID from the IRU instead of typing it.

Table B-4. IRU and noded Error Codes

Error Number	Description	Action
1200	The Registration Did Not Match a License File on the License Server	This occurs during de-registration or update. The customer is trying to de-register/update on the wrong machine. Please check the registration number with the chassis or workstation name to which the license applies, and/or the name of the license server and try again.
1201	Ixia Licensing Daemon Was Not Found on Target Machine	This happens either when the *noded* service is not running or the machine name is incorrect. Please ensure that: • the network is not down • the Node name or IP for the *noded* server is correct • the *noded* service is running on the target machine. Restart *noded* service if it's not running • the proxy setting is correct. See *Proxy Servers and Firewalls* on page B-2 for more information
1202	Invalid Registration Number or Password	This occurs when a bad password is entered in offline mode. Please verify the registration number and password from the email sent to from Ixia, and verify that the IRU is connecting to the right registration server.
1203	Invalid Registration Number	This occurs when the registration number format is incorrect. Please verify the registration number from the email sent from Ixia and try again.
1206	The License Corresponding to This Registration Number Has Not Been Registered Yet	This occurs when the user tries to de-register/upgrade a license without registering it first. Please ensure that the license is registered, and that this is the correct registration number for a license that is to be de-registered, re-registered, or upgraded.
1209	Invalid Password	Please verify your password from the email sent to you earlier and try again, and check that the password was input correctly.

C Licensing Diagnostic Utility

This appendix provides basic information on using the Licensing Diagnostic Utility. This utility helps the user to troubleshoot and understand licensing installation and registration problems.

If licensing difficulties occur, the diagnostic should be run both from the user's PC (where his/her application is run from) and from the license server where licenses are held (usually an Ixia chassis).

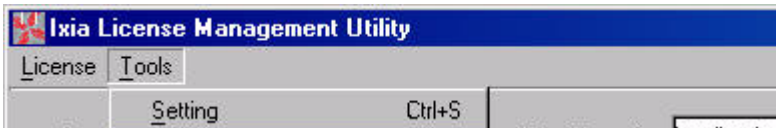
This appendix has the following sections:

- *Opening the Diagnostic Utility* on page C-2
- *Diagnosing License Registration* on page C-6
- *Diagnosing License Procurement* on page C-7

Opening the Diagnostic Utility

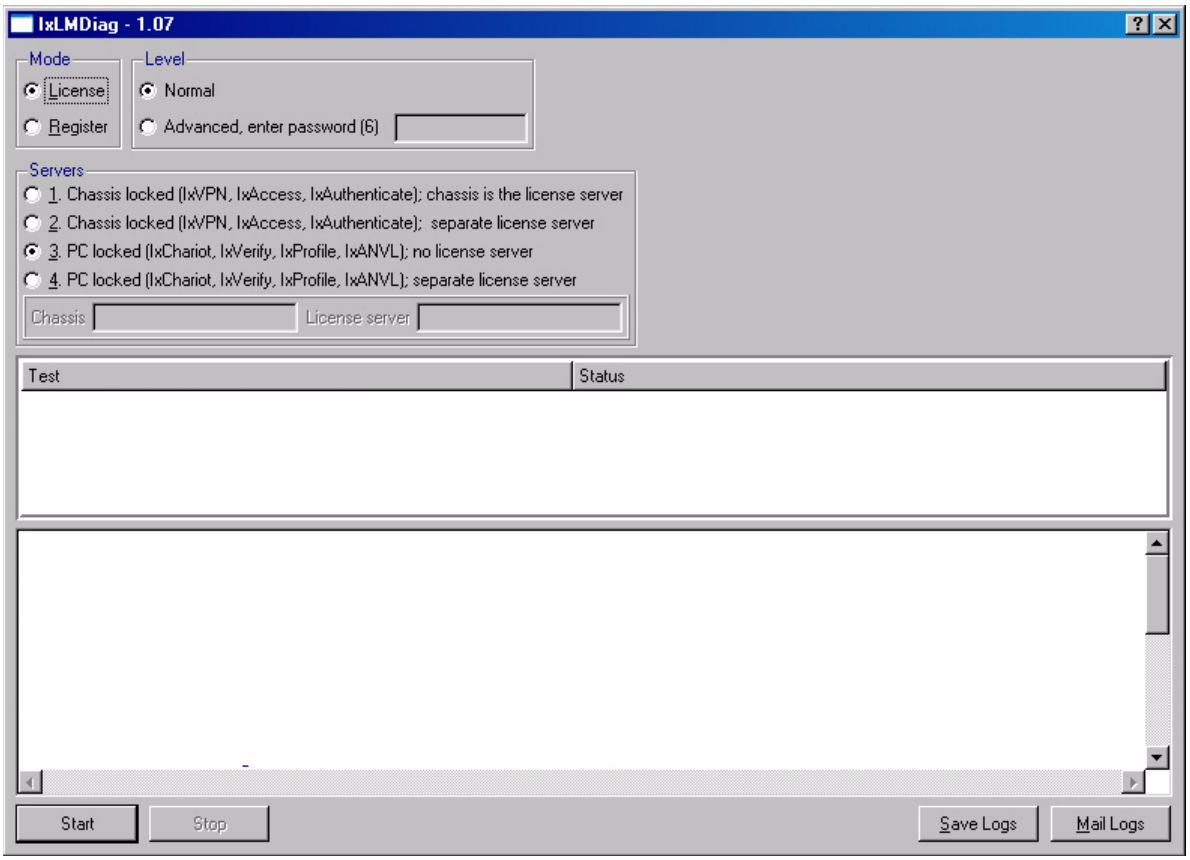
The diagnostic utility helps the user to troubleshoot and understand licensing installation and registration problems. To open this utility, select *Tools* > *Diagnostic* in the menu bar, as shown in Figure C-1 on page C-2:

Figure C-1. Open the Diagnostic Utility



The diagnostic utility main window appears as in Figure C-2 on page C-2

Figure C-2. Diagnostic Utility Main Window



The options available in the diagnostic utility are listed in Table C-1 on page C-3.

Table C-1. Diagnostic Utility Configuration Options

Group	Control	Description
Mode	License	Select this option if you are having trouble obtaining a license. For specific instructions on diagnosing this problem, see *Diagnosing License Procurement* on page C-7.
	Register	Select this option if you are having trouble registering a license. For specific instructions on diagnosing this problem, see *Diagnosing License Registration* on page C-6.
Level	Normal	Only active when *License* mode is selected. Typical user diagnostic level. Use this in all cases.
	Advanced	Only active when *License* mode is selected. Used by Ixia support representatives for on-site licensing help. Requires a password.
Servers	1) Chassis locked; chassis is the license server	Only active when the *License* mode is selected. Select this options when a chassis-locked license is being used and the license server is the same as the chassis being used. Fill in the name of the chassis in the *Chassis* field.
	2) Chassis locked; separate license server	Only active when the *License* mode is selected. Select this options when a chassis-locked license is being used, but the license server is located somewhere other than on the chassis. Fill in the name of the chassis in the *Chassis* field, and the name of the license server in the License server field.
	3) PC locked; no license server	Only active when the *License* mode is selected. Select this options when a PC-locked license is being used, and a license server isn't being used. The license on this computer is checked.
	4) PC locked; separate license server	Only active when the *License* mode is selected. Select this options when a PC-locked license is being used, as well as a separate license server. Fill in name of the license server in the *License* server field.

Table C-1. Diagnostic Utility Configuration Options

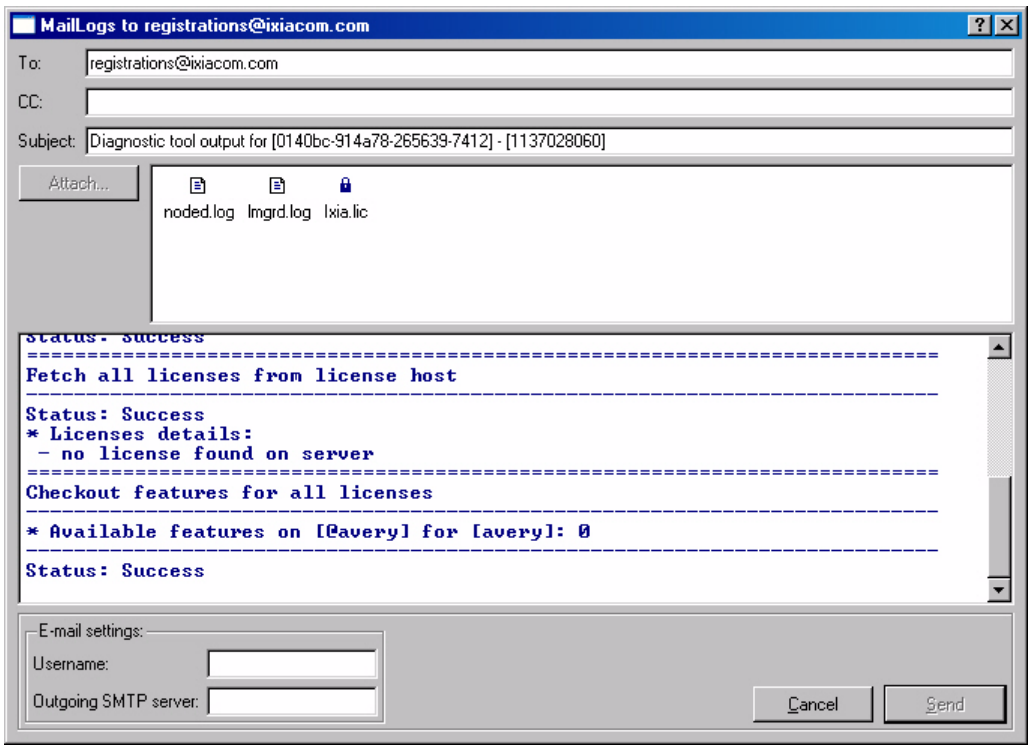
Group	Control	Description
	Chassis	Only active when one of the Chassis locked options is selected in the Server section. Enter the name or IP address of the chassis the to which the chassis is locked.
	License server	Only active when one of the license server options is selected in the Server section. Enter the name or IP address of the license server hosting the license.
Test Windows	Test	Displays which tests have been run.
	Status	Displays the results of the test, either Success or Failure.
Start		Starts the diagnostic process.
Stop		Stops a currently running diagnostic process.
Save Logs		Saves the logs of the last diagnostic run to your local disk.
Mail Logs		Allows the user to mail log files via email. The MailLogs dialog is opened for this purpose. See *Mailing Diagnostic Logs* on page C-5 for more information on how to mail diagnostic logs.

Mailing Diagnostic Logs

After running a diagnostic to determine a licensing problem, it may be necessary to email the information in the logs for the test to Ixia support, or another person within the user's organization.

To send an email with the current diagnostic information, select the *Mail Logs* button in the *Diagnostic* dialog, as described in *Opening the Diagnostic Utility* on page C-2. The *MailLogs* dialog displays, as shown in *MailLogs Dialog* on page C-5. The results of the last diagnostic run are copied to the email.

Figure C-3. MailLogs Dialog



The options in the *MailLogs* dialog are discussed in Table C-2 on page C-5.

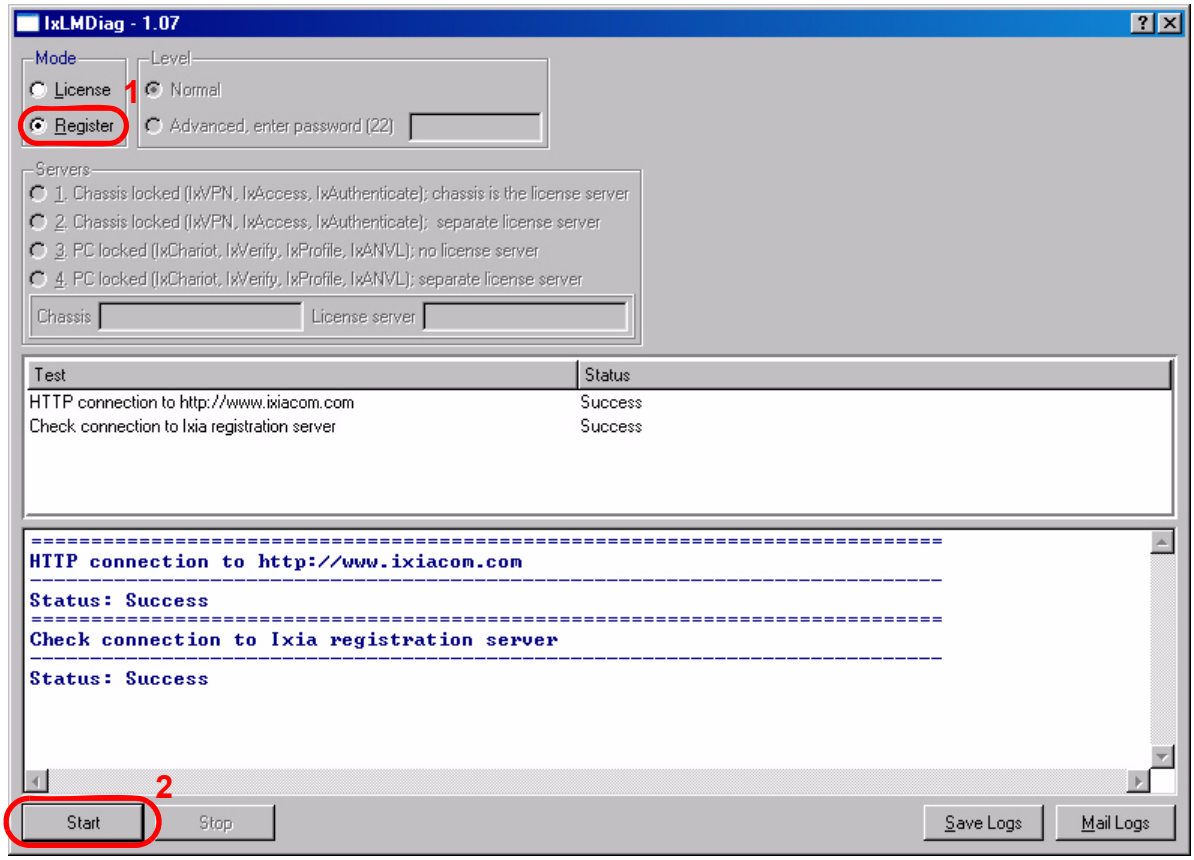
Table C-2.

Control	Description
To/CC/Subject	Standard email options as in any email. The default recipient is always registrations@ixiacom.com
Attach	Use to attach specific files to the email.
Username	Enter a valid username for the mail server.
Outgoing SMTP server	Specify the outgoing SMTP mail server.
Cancel	Cancels the email.
Send	Sends the email to the specified recipients.

Diagnosing License Registration

If a user cannot register their license, select *Register* from the *Mode* section and press *Start*, as shown in Figure C-4 on page C-6:

Figure C-4. Running a Registration Diagnostic



Nominal output is shown in Figure C-4 above.

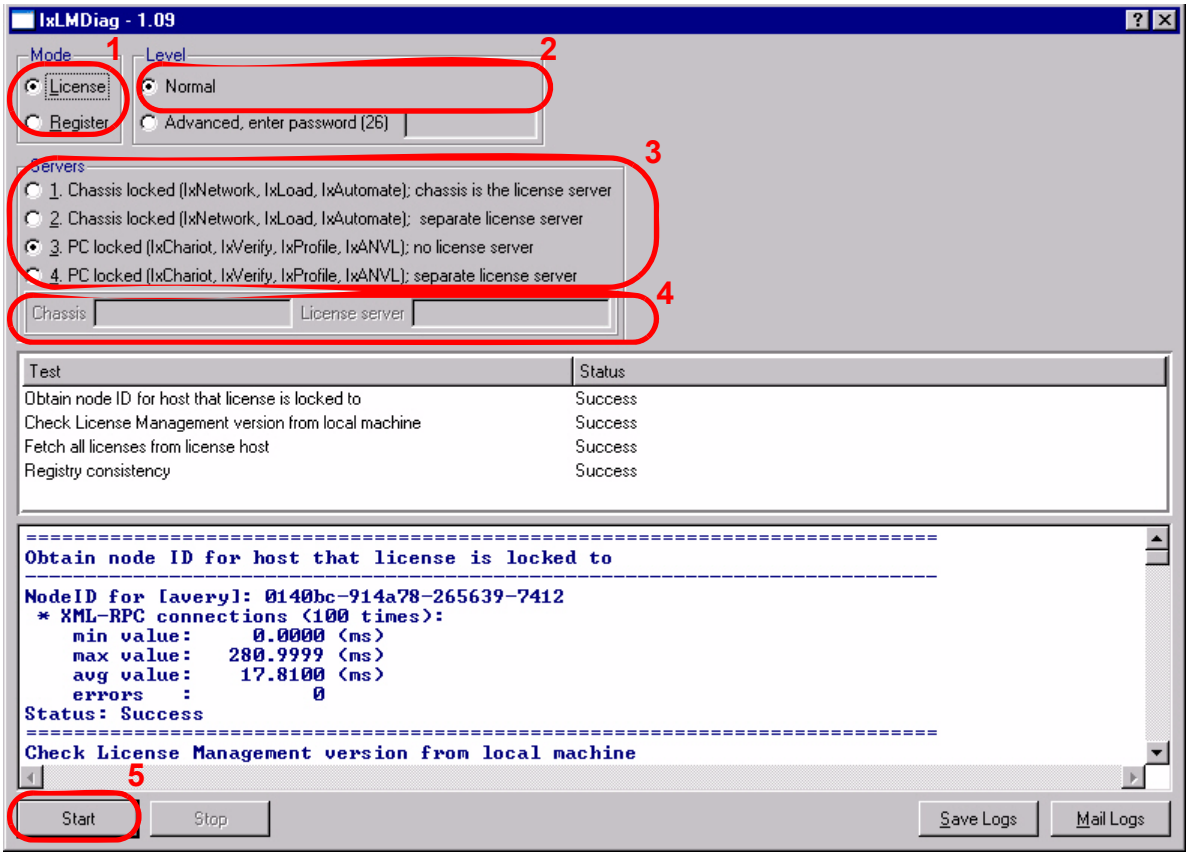
If a user cannot register from a PC, the license server is also a viable location to perform a registration. Make sure to try registration from the license server (chassis). If there is a problem, the diagnostic utility should be run in the same manner from there.

Diagnosing License Procurement

If a license cannot be obtained, the following procedure details how to use the diagnostic utility to find the problem:

1. Select *License* from the *Mode* section.
2. Select *Normal* from the *Level* section.
3. Select the appropriate scenario from the *Server* section, one of:
   - Chassis locked, chassis is the license server
   - Chassis locked, separate license server
   - PC locked, no license server
   - PC locked, separate license server
4. Depending on the server scenario selected, enter the Chassis and License Server name or IP address in the fields provided. These fields are activated only if necessary.
5. Select *Start*. Figure shows an example set-up:

Figure C-5.



For full diagnostics, the utility should be run from the license server as well.

An example of the text of a license diagnostic from the Chassis and from a license server is shown below, at:

- *Chassis Diagnostic Example* on page C-8
- *License Server Diagnostic Example* on page C-11

Chassis Diagnostic Example

```
=========================================================================
Obtain node ID for host that license is locked to
-------------------------------------------------------------------------
NodeID for [avery]: 0140bc-914a78-265639-7412
 * XML-RPC connections (100 times):
    min value:     0.0000 (ms)
    max value:   280.9999 (ms)
    avg value:    17.8100 (ms)
    errors   :          0
Status: Success
=========================================================================
Check License Management version from local machine
-------------------------------------------------------------------------
License Management on this computer
Version: 2.40.0.56
Status: Success
=========================================================================
====
Fetch all licenses from license host
-------------------------------------------------------------------------
----
Status: Success
* Licenses details:
 - License id         : 10024401
 - Registration number: IxiaTemp_NC
 - Product            : Test Conductor
 - Version            : 1.00
 - License Type       : Node Locked
 - License Server     : avery <Matches your license server name>
  * Product features:
    + feature      : 931-1010
    - description  : Aptixia Test Conductor Server, Base Software, Frame
    - expiration   : 10-Feb-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 931-1011
    - description  : AptixiaTest Conductor, Software Option, Console Sea
    - expiration   : 10-Feb-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 931-1012
    - description  : Aptixia Test Conductor SW Opt Pack of 5 Console Sea
    - expiration   : 10-Feb-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>
```

```
-----------------------------------------------------------------------
 - License id         : 10017888
 - Registration number: 500009543NC
 - Product            : IxNetwork
 - Version            : 2007.0120
 - License Type       : Node Locked
 - License Server     : avery <Matches your license server name>
  * Product features:
    + feature      : 895-0100
    - description  : FRAMEWORK, IxRouter/Aptixia IxNetwork
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2004
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, Mult
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2005
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, BGP4
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2006
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, BGP4
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2007
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, BGP4
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2008
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, OSPF
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2009
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, OSPF
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2010
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, IS-I
    - expiration   : 26-Jan-2007 <! Expired>
    - node Id      : 0140bc-914a78-265639-7412 <match your node>

    + feature      : 930-2011
    - description  : IxRouter/Aptixia IxNetwork, Optional Software, IS-I
    - expiration   : 26-Jan-2007 <! Expired>
```

```
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2012
- description  : IxRouter/Aptixia IxNetwork, Optional Software, RIPv
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2013
- description  : IxRouter/Aptixia IxNetwork, Optional Software, RIPn
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2014
- description  : IxRouter/Aptixia IxNetwork, Optional Software, MPLS
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2015
- description  : IxRouter/Aptixia IxNetwork, Optional Software, MPLS
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2017
- description  : IxRouter/Aptixia IxNetwork, Optional Software, STP/
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2018
- description  : Aptixia IxNetwork, Optional Software, MSTP Emulatio
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2019
- description  : Aptixia IxNetwork, Optional Software, PVST+/RPVST+
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2020
- description  : IxNetwork, Optional Software, EIGRP Emulation; REQU
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2021
- description  : Aptixia IxNetwork, Optional Software, Application T
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>

+ feature      : 930-2022
- description  : Aptixia IxNetwork, Optional Software, RFC 2544 and
- expiration   : 26-Jan-2007 <! Expired>
- node Id      : 0140bc-914a78-265639-7412 <match your node>
```

```
-----------------------------------------------------------------------
=========================================================================
Registry consistency
-----------------------------------------------------------------------
* Registry consistency details:
   - Valid registry checksum
Status: Success
```

License Server Diagnostic Example

```
=========================================================
Obtain node ID for host that license is locked to
---------------------------------------------------------
NodeID for [localhost]: 011cf9-00cb8c-f34ce9-c000
  * XML-RPC connections (100 times):
    min value:    20.0000 (ms)
    max value:    89.9999 (ms)
    avg value:    49.1700 (ms)
    errors   :          0
Status: Success
=========================================================
Check License Management version from local machine
---------------------------------------------------------
License Management on this computer
Version: 2.0.9.11
Status: Success
=========================================================
Fetch all licenses from license host
---------------------------------------------------------
Status: Success
* Licenses details:
  - License id         : 10004120
  - Registration number: 500003203NC
  - Product            : IXACCESS
  - Version            : 2.10
  - License Type       : Node Locked
    * Product features:
      + feature      : IXACCESS
      - description  : PPPoE EMULATION SW
      - expiration   : permanent
      - node Id      : 011cf9-00cb8c-f34ce9-c000 <match
                              your node>


---------------------------------------------------------
  - License id         : 10002264
  - Registration number: 500001470NS
  - Product            : IxChariot-NL
  - Version            : 6.10
  - License Type       : Node Locked
    * Product features:
      + feature      : 920-FW-NL
      - description  : IxChariot Console,VOIP, IPV6 (Node
                               Locked)
      - expiration   : permanent
      - node Id      : 011cf9-00cb8c-f34ce9-c000 <match
                               your node>

      + feature      : 920-PAR5000-NL
      - description  : IxChariot - 5000 Pairs (Node
                               Locked)
      - expiration   : permanent
      - node Id      : 011cf9-00cb8c-f34ce9-c000 <match
                               your node>
```

```
-------------------------------------------------------
  - License id         : 10002800
  - Registration number: 500002318NC
  - Product            : IXVPN
  - Version            : 2.10
  - License Type       : Node Locked
    * Product features:
      + feature      : IXVPN
      - description  : Application for Testing IPSec
                               Devices
      - expiration   : permanent
      - node Id      : 011cf9-00cb8c-f34ce9-c000 <match
                               your node>

      + feature      : IXVPN_MNL_IKE
      - description  : Manual IKE for IxVPN
      - expiration   : permanent
      - node Id      : 011cf9-00cb8c-f34ce9-c000 <match
                               your node>


-------------------------------------------------------
=======================================================
Checkout features for all licenses
-------------------------------------------------------
* Available features on [@localhost] for [localhost]: 8
  + Feature name: ixchariot_ns
  | - Version     : 6.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: chr_ip6_ns
  | - Version     : 6.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: chr_voip_ns
  | - Version     : 6.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: chr_video_ns
  | - Version     : 6.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: chr_pairs_ns
  | - Version     : 6.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: IXVPN
  | - Version     : 2.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: IXVPN_MNL_IKE
  | - Version     : 2.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
  + Feature name: IXACCESS
  | - Version     : 2.10
  | - Status      : Ok
  | - Exp. date   : 1-jan-0
  | - Err. message:
-------------------------------------------------------
Status: Success
```

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